

2021-Ph-H1-Q11

Section: Our Dynamic Universe

Topic: The Expanding Universe

Which graph best shows the relationship between recessional velocity and distance for galaxies?

Solution:

According to **Hubble's Law**:

$$v = H_0 d$$

This is a straight-line graph through the origin.

Answer: C

Guidance for Students:

Recessional velocity and distance are directly proportional — the further away a galaxy, the faster it moves.

Revision Tip:

Hubble's Law = straight-line graph

Know the formula:

$$v = H_0 d$$

Units: km/s and Mpc